A.5.2 Zero Ablations: Full NLP ablation experiments

Llama-3-8B						
Task	Shot	Full model	1% ind. heads	1% rnd. heads	3% ind. heads	3% rnd. heads
BoolQ	0	77.8	76.2 (-1.6)	72.7 (-5.1)	75.2 (-2.6)	71.9 (-5.9)
	5	79.8	77.1 (-2.7)	76.4 (-3.4)	73.0 (-6.8)	77.2 (-2.6)
	10	79.6	77.7 (-1.9)	77.1 (-2.5)	70.7 (-8.9)	77.3 (-2.3)
SUL	10	46.3	-	-	-	-
ETHOS	0	70.0	72.0 (+2.0)	68.6 (-1.4)	70.0	62.0 (-8.0)
	5	82.5	82.5	79.5 (-3.0)	81.2 (-1.3)	75.3 (-7.2)
	10	80.2	80.2	80.9 (+0.7)	81.1 (-1.1)	77.5 (-2.7)
SUL	10	70.3	52.8 (-17.5)	66.8 (-3.5)	57.9 (-12.4)	63.8 (-6.5)
RTE	0	71.1	67.5 (-3.6)	70.3 (-0.8)	67.9 (-3.2)	67.5 (-3.6)
	5	79.5	75.2 (-4.3)	79.0 (-0.5)	71.6 (-7.9)	78.1 (-1.4)
	10	79.4	74.2 (-5.2)	78.7 (-0.7)	68.6 (-10.8)	78.4 (-1.0)
SUL	10	52.4	-	-	-	-
SST-2	0	92.4	92.1 (-0.3)	92.7 (+0.3)	92.2 (-0.2)	91.9 (-0.5)
	5	93.9	92.4 (-1.5)	94.3 (+0.4)	93.0 (-0.9)	93.9
	10	94.6	92.9 (-1.7)	94.7 (+0.1)	93.2 (-1.4)	94.5 (-0.1)
SUL	10	94.0	72.8 (-21.2)	93.6 (-0.4)	71.8 (-22.2)	92.6 (-1.4)
SUBJ	0	52.6	53.0 (+0.4)	54.0 (+1.4)	52.4 (-0.2)	53.2 (+0.6)
	5	58.1	53.5 (-4.6)	58.8 (+0.7)	56.3 (-1.8)	56.5 (-1.6)
	10	74.7	62.4 (-12.3)	74.6 (-0.1)	62.6 (-12.1)	71.9 (-2.8)
SUL	10	69.8	49.9 (-19.9)	67.0 (-2.8)	53.1 (-16.7)	70.4 (+0.6)
WIC	0	51.4	51.4	51.0 (-0.4)	50.9 (-0.5)	51.7 (+0.3)
	5	59.6	58.2 (-1.4)	59.0 (-0.6)	56.3 (-3.3)	56.8 (-2.8)
	10	61.6	56.8 (-4.8)	59.6 (-2.0)	56.2 (-5.4)	57.0 (-4.6)
SUL	10	52.6	-	-	-	-
InternLM2-20B						
Task	Shot	Full model	1% ind. heads	1% rnd. heads	3% ind. heads	3% rnd. heads
BoolQ	0	88.6	88.6	88.5 (-0.1)	87.7 (-0.9)	86.9 (-1.7)
	5	88.5	88.5	88.6 (+0.1)	88.2 (-0.3)	87.5 (-1.0)
	10	88.3	88.3	88.2 (-0.1)	87.9 (-0.4)	87.1 (-1.3)
SUL	10	80.7	72.4 (-8.3)	80.9 (+0.2)	64.9 (-15.8)	77.0 (-3.7)
ETHOS	0	82.0	82.5 (+0.5)	82.5 (+0.5)	82.0	81.5 (-0.5)
	5	84.7	81.3 (-3.4)	84.2 (-0.5)	81.8 (-2.9)	82.6 (-2.1)
	10	82.6	81.0 (-1.6)	82.9 (+0.3)	82.1 (-0.5)	82.8 (+0.2)
SUL	10	82.5	74.8 (-7.7)	81.3 (-1.2)	68.2 (-14.3)	74.4 (-8.1)
RTE	0	81.2	80.1 (-1.1)	80.6 (-0.6)	80.5 (-0.7)	79.9 (-1.3)
	5	84.7	82.3 (-2.4)	84.4 (-0.3)	81.4 (-3.3)	84.4 (-0.3)
	10	87.2	84.0 (-3.2)	86.8 (-0.4)	82.3 (-4.9)	86.4 (-0.8)
SUL	10	80.1	62.9 (-17.2)	81.4 (+1.3)	56.4 (-23.7)	76.9 (-3.2)
SST-2	0	96.0	96.0	95.7 (-0.3)	95.6 (-0.4)	95.7 (-0.3)
	5	95.1	95.1	95.0 (-0.1)	94.2 (-0.9)	94.9 (-0.2)
	10	95.3	95.1 (-0.2)	95.2 (-0.1)	94.3 (-1.0)	95.2 (-0.1)
SUL	10	94.1	93.6 (-0.5)	94.1	92.2 (-1.9)	93.6 (-0.5)
SUBJ	0	61.1	61.7 (+0.6)	61.6 (+0.5)	62.6 (+1.5)	59.9 (-1.2)
	5	75.4	69.6 (-5.8)	75.0 (-0.4)	70.0 (-5.4)	73.4 (-2.0)
	10	81.4	75.1 (-6.3)	81.4	71.3 (-10.1)	78.8 (-2.6)
SUL	10	64.9	53.4 (-11.5)	65.1 (+0.2)	52.2 (-12.7)	61.3 (-3.6)
WIC	0	60.8	60.7 (-0.2)	60.8	59.9 (-0.9)	60.1 (-0.7)
	5	68.6	67.2 (-1.4)	68.7 (+0.1)	61.9 (-6.7)	68.9 (+0.3)
	10	68.9	67.0 (-1.9)	68.6 (-0.3)	63.5 (-5.4)	68.3 (-0.6)
SUL	10	57.2	54.8 (-2.4)	57.7 (+0.5)	56.6 (-0.6)	54.9 (-2.3)
WSC	0	70.2	67.3 (-2.9)	69.6 (-0.6)	68.3 (-1.9)	70.8 (+0.6)
	5	73.1	68.0 (-5.1)	73.0 (-0.1)	69.6 (-3.5)	73.1
	10	75.3	69.9 (-5.4)	75.2 (-0.1)	68.6 (-1.3)	71.9 (-3.4)
SUL	10	43.3	-	-	-	-

Table 5: LLama-3-8B and InternLM2-20B ablation experiments on the NLP tasks. Columns labelled "1% ind. heads" and "3% ind. heads" show the results from fully ablating 1% and 3% of heads with the highest prefix scores, respectively. Columns "1% rnd. heads" and "3% rnd. heads" illustrate the effects of randomly ablating 1% and 3% of all heads in the model. The "SUL" row denotes settings using semantically unrelated labels. Performance differences due to the ablation, compared to the full model, are indicated in parentheses.

A.5.3 Zero Ablations: NLP Figures

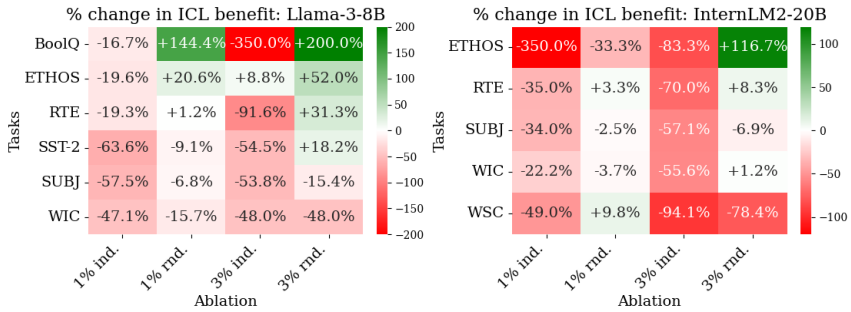

Figure 9: Change in ICL benefit for Llama-3-8B (left) and InternLM2-20B (right), due to head ablations when compared to that of the full model. "1% ind." and "3% ind." denote ablating the top respective percentage of induction heads. "1% rnd." and "3% rnd." denote randomly ablating the respective percentage of all heads in the model.

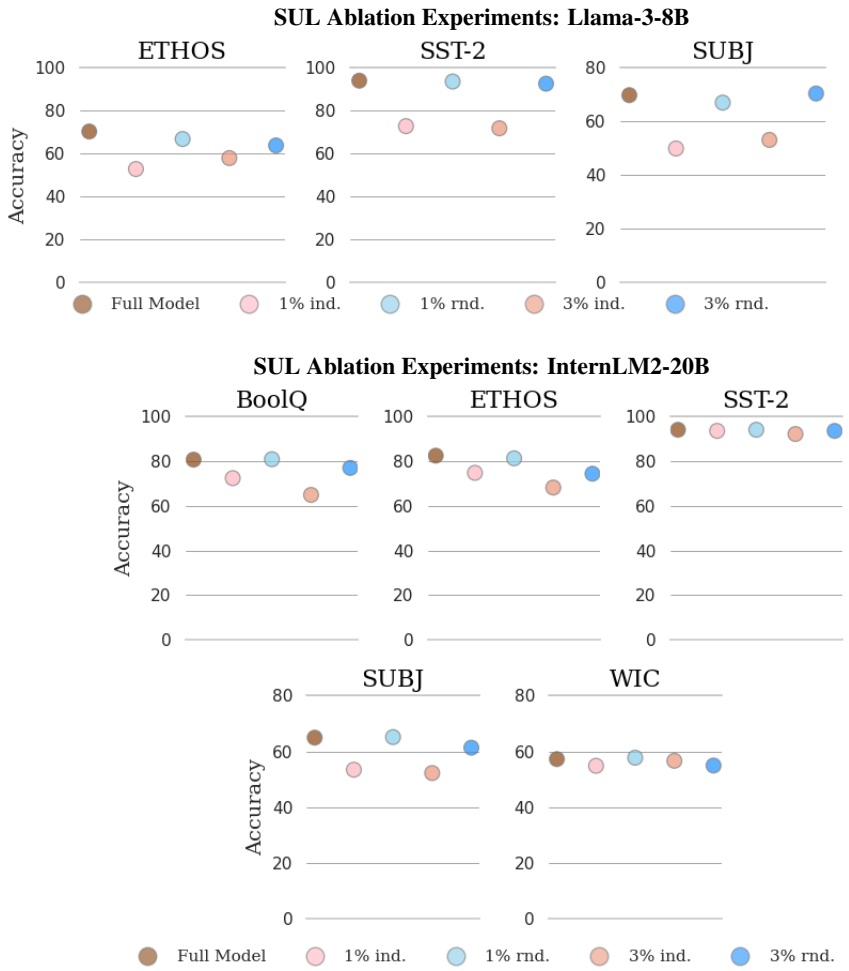

Figure 10: SUL ablation experiments for Llama-3-8B (top) and InternLM2-20B (bottom) on NLP tasks. "1% ind." and "3% ind." denote ablating the top respective percentage of induction heads. "1% rnd." and "3% rnd." denote randomly ablating the respective percentage of all heads in the model.

A.6 Attention Patterns

A.6.1 WordSeq1 (Binary)

Figure 11: Attention pattern of token 62 in head 22 of layer 2 in Llama-3-8B. Example 1/5.

Figure 12: Attention pattern of token 64 in head 22 of layer 2 in Llama-3-8B. Plot 1/5.

Figure 13: Attention pattern of token 62 in head 1 of layer 8 in Llama-3-8B. Example 1/5.

Figure 14: Attention pattern of token 64 in head 1 of layer 8 in Llama-3-8B. Plot 1/5.